The microscopic clocks in bacteria

Introduction

- Bacteria make up >10% of all living things
- Non photosynthetic soil bacteria have internal clocks that synchronise with day/night cycles
- This can be used in precision timing of antibiotic use or to bioengineer smarter gut and soil microbiomes

Wait... what are circadian rhythms?
They are internal timing mechanisms enabling living organisms to cope with major changes in environment (e.g. daylight, temperature) that occur from day to night.

One example of this is jet lag, where our internal clocks are temporarily mismatched before aligning to the new cycle of light and dark at our travel destination!

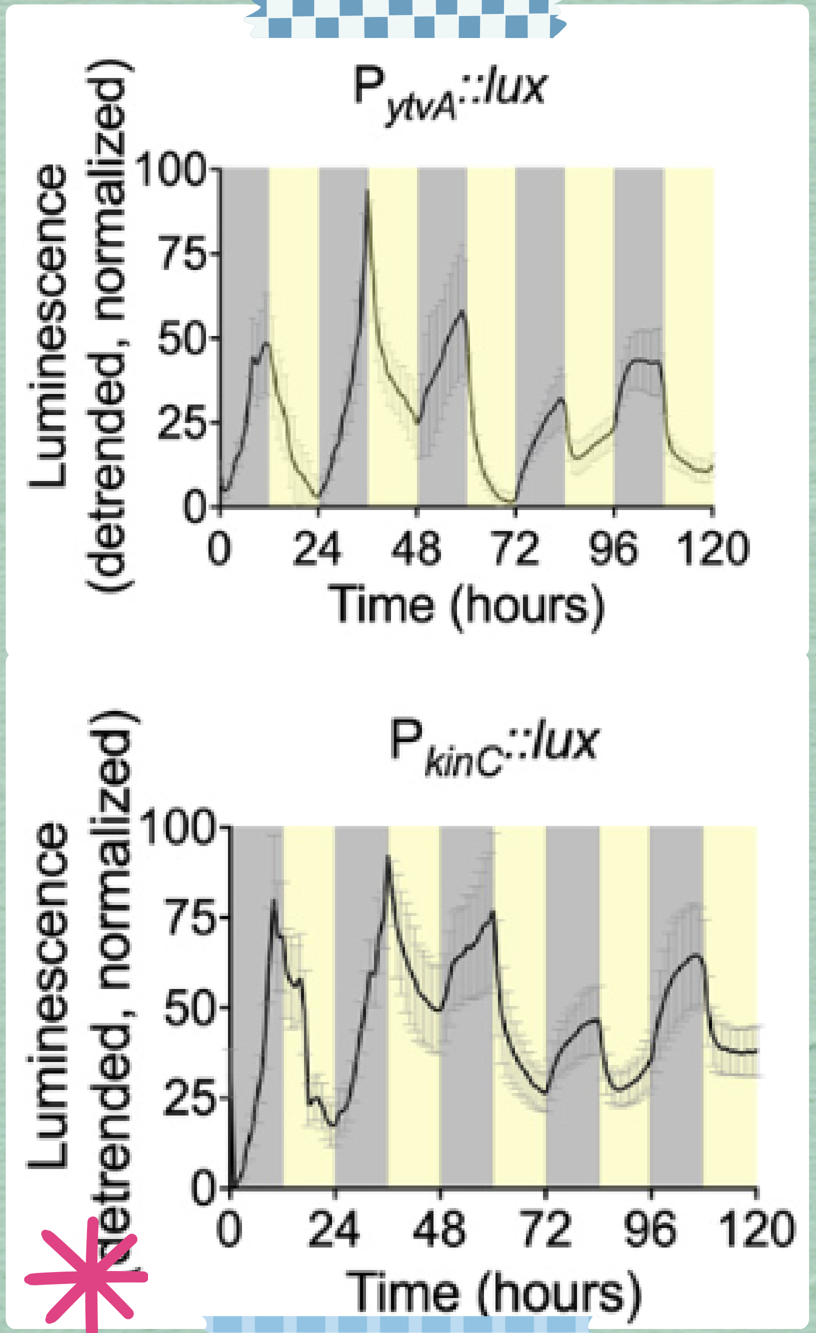
How does it work?

Researchers detected circadian rhythms in Bacillus subtilis by adding bioluminescent luciferase enzyme to visualise gene expression.

- In constant 12 hour day/night cycles:
- ytvA, a blue light photoreceptor and KinC, a biofilm and spore formation enzyme showed fluctuating expression levels

- Expression of both genes adjusted to the light/dark cycle, increasing during the dark and decreasing in the light
- A cycle was still observed in constant darkness

- This showed 1) reversibility and 2) acclimatisation, common features of circadian rhythms.
- Expression levels were affected by temperature and light intensity



Why is this research important?

Understanding properties of bacterial circadian clocks may:

- Further knowledge of how microbiomes are formed
- Assess antibiotic efficacy at certain times of the day to better utilise them against resistant bacteria
- Assist nutrient exchange, plant development and defense against pathogenic microbes in crops

References

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